Can comets deliver prebiotic molecules to rocky exoplanets?

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Introduction

Comets have been suggested as potentially important sources of prebiotic delivery to the early-Earth [1, 2]. Very low-velocity impacts are required for prebiotic survival, due to the high temperatures during impacts [3]. Comets contain large amounts of the prebiotic feedstock molecule hydrogen cyanide (HCN) [4], which has a strong carbon-nitrogen triple bond, so is more durable to high temperatures. Cometary delivery is thought to have been able to deliver high concentrations of HCN to local environments [5]. **In this work, we consider the potential of cometary impacts to deliver complex organic molecules, and the prebiotic building blocks required for life to rocky exoplanets.**

Idealised Planetary Architecture

Inspired by observations of tightly-packed planetary systems we consider a **chain of equal-mass ($M_{\rm Earth}$), co-planar planets** around a central star ($M_* \in [0.1, 2]\, M_{\rm Sun}$). We fix the inner planet in the centre of the star's habitable zone. We make two further assumptions:

- The planets are equally spaced in terms of their mutual Hill radius.

$$a_{i+1} = a_i + \Delta R_{H_{i,i+1}} \tag{1}$$

- Volatile-rich comets are scattered from beyond the snow-line, and so the number of planets is chosen to span the distance between the snow-line and habitable zone.

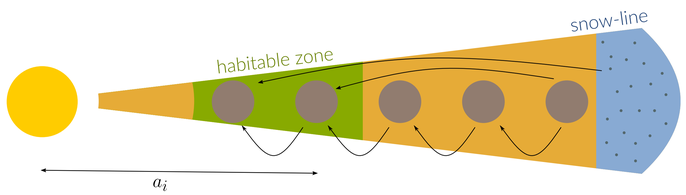

Figure 1. Schematic diagram of the idealised planetary system considered in this work. Low velocity impacts follow the lower arrows, demonstrating how comets can be passed along the chain of planets.

Analytical Model

Simulations have shown that scattering by multiple planets can be modelled by a series of three-body interactions [6]. We separate the dynamics of comets into two separate regions - inside and outside of the planet's Hill radius,

- **Outside:** the comet's dynamics are determined by its Tisserand parameter, $\mathcal{T}$. The relative velocity to the planet will be $v_\infty = v_{\rm pl}\sqrt{3-\mathcal{T}}$.
- **Inside:** the comet's dynamics are set by the scattering planet, and follows a hyperbolic orbit with the scattering planet at the focus. The impact velocity is therefore $v_{\rm imp}^2 = v_{\rm esc}^2 + v_\infty^2$.

The minimum impact velocity (maximum Tisserand parameter) is determined by the architecture, corresponding to comets scattered by the nearest planet. These can be calculated analytically,

$$\mathcal{T}_{\rm max} = (1-\Delta X) + 2\sqrt{1+\Delta X}, \tag{2}$$

$$v_{\rm imp,min}^2 = v_{\rm esc}^2 + \left(\frac{GM_*}{d_{\rm HZ}}\right)\left(2+\Delta X - 2\sqrt{1+\Delta X}\right), \tag{3}$$

where $d_{\rm HZ} = d_{\rm HZ}(M_*)$ is the radial location of the habitable zone.

Analytical and Numerical Results

We find that there are three key parameters that determine the minimum impact velocity onto a planet, **the planet's escape velocity, the planet's orbital velocity and the particle's orbit relative to the planet**, as characterised by the Tisserand parameter (see figures 2 and 4).

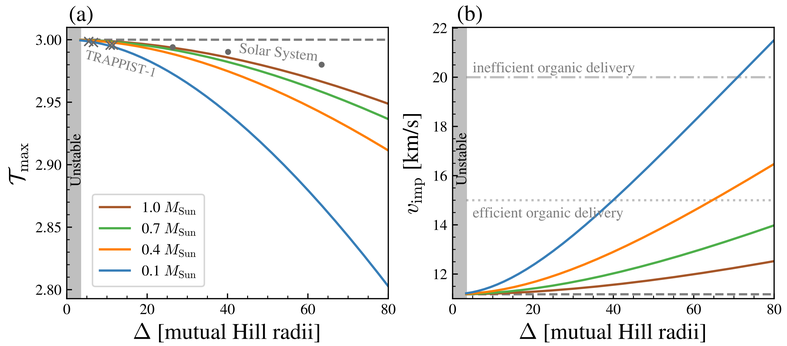

Figure 2. Analytical predictions for the maximum Tisserand parameter **(left)** and minimum impact velocity **(right)** onto an Earth-like planet in the habitable zone. Rough velocity thresholds, based off values for HCN survival [5] are added for reference.

N-body test-particle simulations of these idealised systems show **there is a much larger fraction of very low-velocity impacts around high-mass stars and in more tightly-packed systems.** It is not only the minimum impact velocity that changes!

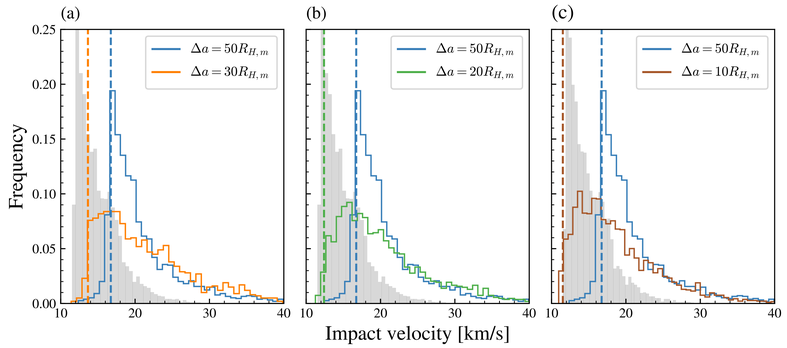

Figure 3. Numerical impact velocity distributions onto an Earth-like planet in the habitable zone of a low-mass star. For comparison, the velocity distribution for a planet around a G-type star ($1M_{\rm Sun}, \Delta a = 50R_{H,m}$) is included in grey. Dashed lines indicate the analytical predictions shown in figure 2.

For comets scattered directly from the snow-line onto the planet, we see that **very low impact velocities are impossible around low-mass stars**, which could prevent successful cometary delivery in specific scenarios.

There is a much larger spread in impact velocities around lower-mass stars, which remains almost unchanged by the planetary architecture. The increased fraction of high-velocity impactors may be prejudicial to the development of life, and lead to increased atmospheric erosion.

Observational Outlook

The next generation of great observatories (e.g. HabEx, PLATO, LIFE) aim to detect and characterise a census of potentially habitable planets. This work predicts that if cometary delivery of prebiotic molecules is important for the origins of life, **there will be a correlation between the presence of biosignatures and the following exoplanet populations,**

- **Exoplanets with low escape velocities.** A strict upper bound around 25 km/s would be a very strong indicator that cometary delivery is important.
- **Exoplanets orbiting high-mass stars** ($M_* \gtrsim 1M_{\rm Sun}$), as Keplerian orbital velocities scale roughly as $M_*^{-1/2}$.
- **Exoplanets in tightly-packed planetary systems**, for which comets are delivered on low eccentricity orbits.

These populations of exoplanets correspond to the labelled regions of low impact velocity seen below in figure 4, for which planetary separation and stellar-mass are equally important.

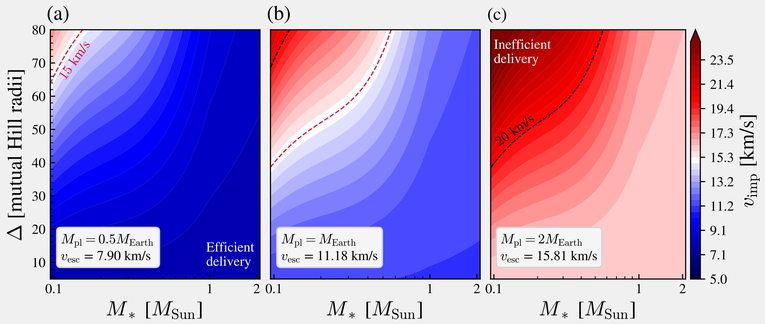

Figure 4. Contour plots of the minimum impact velocity onto habitable Earth-like planets, across a wide range of Δ-M_* parameter space. The planet's escape velocity is changed between the three panels.

In summary, these results highlight that **the successful cometary delivery of prebiotic molecules is very sensitive to planet's bulk composition, surrounding planetary architecture and stellar-mass.**

Acknowledgements

Simulations in this work made use of the REBOUND code (specifically the MERCURIUS integrator) which can be downloaded freely at `https://github.com/hannorein/rebound`.

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